



Overview of Research Findings on 24-Hour Movement Guidelines and Overweight/Obesity in School-Aged Children and Adolescents

Overview of Research Findings

Research examining adherence to 24-hour Movement Guidelines and overweight/obesity risk in school-aged children and adolescents (ages 5-18, excluding preschoolers) reveals substantial evidence supporting a protective association between meeting these integrated guidelines and reduced obesity risk. Based on systematic review data and individual studies published through May 2023, the evidence demonstrates clear patterns:

Studies Finding a Correlation:

Approximately 8 major individual studies and multiple meta-analyses documented significant associations between adherence to 24-hour Movement Guidelines and lower overweight/obesity risk in school-aged children and adolescents:

- **Roman-Viñas et al. (2016)** across 12 countries found that children meeting all three recommendations had substantially lower odds of obesity (OR = 0.28, 95% CI 0.18-0.45) compared to those meeting none^[1]
- **Carson et al. (2017)** in Canada demonstrated a clear dose-response relationship, where meeting fewer recommendations was progressively associated with higher BMI z-scores and waist circumference^[2]
- **Katzmarzyk et al. (2017)** in the USA showed significant associations between the number of guidelines met and multiple obesity indicators including BMI, visceral adipose tissue, and cardiometabolic risk factors^[3]
- **Zhu et al. (2020)** in a US nationally representative sample of 30,478 adolescents found that meeting zero guidelines versus all three was associated with 4.25-fold higher odds of obesity (95% CI 2.87-6.31)^[4]
- **Tanaka et al. (2020)** in Japan demonstrated that multiple guideline combinations were associated with lower odds of overweight/obesity, with the most consistent findings for screen time recommendations^[5]
- **Chen et al. (2020/2021)** in China examined 114,072 children and adolescents, finding that those meeting guidelines showed significantly lower odds of overweight/obesity, with stronger effects in younger age groups^[6]
- **Chemtob et al. (2021)** demonstrated longitudinal associations, where children meeting fewer guideline components in childhood had substantially higher adiposity measures 2 and 7 years later, with a 1.66 standard deviation difference in BMI z-score^[7]

- **Leppänen et al. (2022)** in Finland showed that meeting physical activity guidelines, particularly in combination with other recommendations, was longitudinally associated with lower cardiometabolic risk and waist circumference^[8]
- **López-Gil et al. (2023)** meta-analysis found cross-sectional associations between meeting all recommendations and lower obesity indicators (OR = 0.65 for overweight/obesity, 95% CI: 0.56-0.76; OR = 0.28 for obesity alone, 95% CI: 0.16-0.50), though no longitudinal associations emerged^[9]
- **Nagata et al. (2023)** demonstrated that the combination of low screen time and high physical activity was most protective, with high screen time substantially increasing obesity risk regardless of activity level^[10]
- **García-Hermoso et al. (2023)** in a 14-year longitudinal follow-up found that meeting screen time and physical activity recommendations during adolescence was associated with lower risk of abdominal obesity in adulthood (IRR = 0.86, 95% CI 0.67-0.97)^[11]

Studies Finding No Correlation:

Systematic review evidence (Marques et al., 2023) identified that among school-aged children and adolescents studies, approximately 6 out of 15 found no significant associations between meeting 24-hour movement guidelines and obesity/overweight indicators, and 1 study showed mixed results depending on age stratification. However, these studies were less frequently represented in the literature compared to those reporting positive associations.^[12]

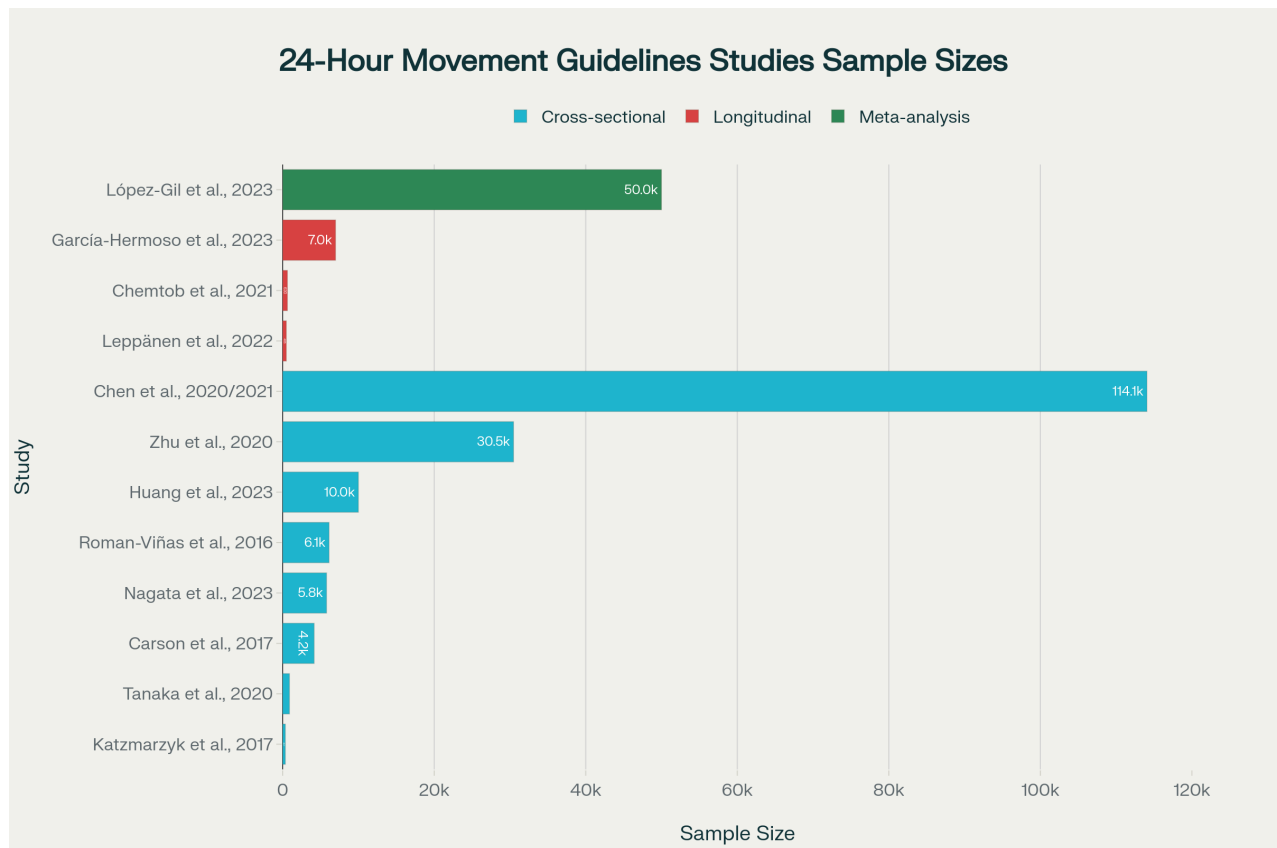
Key Specific Conclusions from Studies Finding Correlations:

The research consistently demonstrated that:

1. **Dose-response relationships:** More guidelines met was associated with progressively lower obesity risk^{[2] [3]}
2. **Physical activity primacy:** The moderate-to-vigorous physical activity (MVPA) component was most consistently associated with lower obesity risk, even when analyzed independently^{[4] [1]}
3. **Combined effects most protective:** Meeting all three recommendations (MVPA ≥60 min/day, screen time ≤2 h/day, adequate sleep) consistently showed the strongest protective effects, though meeting individual recommendations also demonstrated protective associations^{[7] [10]}
4. **Longitudinal evidence:** Studies with follow-up periods demonstrated that non-adherence in earlier childhood predicted higher adiposity in later years^{[11] [8] [7]}
5. **Population consistency:** Protective associations were observed across diverse geographic regions (USA, Canada, China, Japan, Finland, Australia) and 12-country international samples, suggesting generalizability^{[6] [10] [1] [4]}
6. **Measurement variations:** Associations were consistent across different obesity indicators including BMI, BMI z-score, waist circumference, body fat percentage, and visceral adiposity^{[3] [1] [2]}

Important Limitations:

- The López-Gil meta-analysis found cross-sectional associations but notably no significant longitudinal associations^[9]
- Effect sizes, while statistically significant, were often modest in magnitude
- Very low adherence rates globally (typically 5-17% meeting all three recommendations) limited ability to study high adherence populations^{[1] [4] [6]}



Summary Table of Research Studies Examining 24-Hour Movement Guidelines and Overweight/Obesity in School-Aged Children and Adolescents (Published Before May 2023)



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